MEDICAL INSURANCE EXPENDITURE ANALYSIS

An Empirical Diagnostic and Modeling Study

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ABSTRACT

This project models individual medical insurance charges for 1,338 observations and evaluates how well a conventional multiple linear regression model represents the relationships in the data. The baseline ordinary least squares (OLS) model achieved a training R^2 of 0.7420 and a test R^2 of 0.7838. Residual diagnostics indicated heavy-tailed residuals and non-random patterns in the residuals-versus-fitted plot, suggesting that a main-effects-only linear specification did not capture all of the structure in the data. Guided by exploratory analysis, squared terms for age and BMI and interaction terms between smoking status and both age and BMI were added, then an expanded linear model was compared with a Random Forest Regressor using five-fold cross-validation. The expanded linear model achieved a mean cross-validation R^2 of 0.8362, while the Random Forest achieved 0.8569. On the held-out test set, the final Random Forest model reached an R^2 of 0.8632 and an RMSE of $4,608.29. These results illustrate how diagnostic analysis and feature engineering can improve predictive performance, while highlighting the importance of evaluating models on data not used for fitting.

Keywords: medical insurance charges; multiple linear regression; residual diagnostics; feature engineering; interaction terms; Random Forest

1. PROJECT MOTIVATION AND OBJECTIVES

Medical insurance charges vary across demographic and lifestyle characteristics. A useful predictive model should capture meaningful patterns in these characteristics while being assessed for the limitations of its assumptions. This project uses a dataset of 1,338 individuals to examine the predictive performance of multiple linear regression, inspect its residuals, and assess whether feature engineering and a tree-based ensemble model improve prediction on unseen observations.

The analysis addresses three questions:

- How well does a main-effects multiple linear regression model predict medical insurance charges?
- What patterns are revealed by residual diagnostics, and how can those patterns inform model development?
- Do engineered features and a Random Forest Regressor improve predictive performance on held-out data?

2. DATASET AND VARIABLES

The dataset contains 1,338 observations, six candidate predictors, and a continuous outcome representing medical charges. The reported variables and summary statistics are listed below.

Variable	Description / summary
Age	18–64 years; mean 39.21; standard deviation 14.05.
Body mass index (BMI)	15.96–53.13; mean 30.66; standard deviation 6.10.
Children	Discrete count from 0 to 5; mean 1.09.
Smoking status	1,064 non-smokers and 274 smokers.
Sex	676 male and 662 female observations.
Region	Four geographic categories.
Medical charges	Mean $13,270.42; median $9,382.03; maximum $63,770.43.

Categorical predictors were dummy-encoded using reference categories. Medical charges show a substantial difference between the mean and median, consistent with a right-skewed outcome distribution.

3. BASELINE MULTIPLE LINEAR REGRESSION

The observations were split into an 80% training set (n = 1,070) and a 20% held-out test set (n = 268). A multiple linear regression model was fitted with statsmodels to establish a baseline. The model included age, BMI, number of children, smoking status, sex, and regional terms.

Baseline model

$$\hat{Y} = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{BMI}) + \beta_3(\text{Children}) + \beta_4(\text{Smoker}) + \beta_5(\text{Sex}) + \text{Regional terms}$$

The fitted model estimated a large positive coefficient for smoking status (approximately 23,660), while age and BMI were also positively associated with charges in the fitted model. Sex and region were reported as statistically insignificant at the 10% level. These are model-based associations and should not be interpreted as causal effects.

Baseline fit and collinearity checks

Metric	Reported result
Training R^2	0.7420
Adjusted training R^2	0.7400
F-statistic	380.7 ($p < 0.001$)
Baseline test R^2	0.7838
Variance inflation factors	All reported values below 1.70; examples: age 1.02, BMI 1.11, smoker 1.01.

Residual diagnostics

The residual analysis identified two notable patterns:

- **Heavy-tailed residuals:** The KDE and normal Q–Q plots showed departures from normality at the extremes. The reported Jarque–Bera statistic was 612.87 ($p < 0.001$). This indicates that the residual distribution departs from normality; it does not, by itself, establish that predictions are biased.
- **Residual structure and possible heteroscedasticity:** The residuals-versus-fitted plot showed non-random patterns rather than a uniform scatter. This suggests that the main-effects-only specification may be missing structure, and that the error variance may not be constant across fitted values. The plot motivates further modeling but does not alone prove that specific interactions caused the pattern.

4. FEATURE ENGINEERING AND MODEL COMPARISON

Exploratory plots suggested that the relationship between age or BMI and medical charges may differ by smoking status. To represent potential curvature and non-additive relationships, the feature set was expanded with age squared, BMI squared, age-by-smoker, and BMI-by-smoker terms.

Models were compared using five-fold cross-validation on the training data. The reported mean cross-validation R^2 values are shown below.

Model	Specification	Mean 5-fold CV R^2
Expanded linear regression	Baseline predictors plus engineered squared and interaction terms	0.8362
Random Forest Regressor	Ensemble of trees; reported max_depth = 5	0.8569

The Random Forest had the higher mean cross-validation R^2 of the two evaluated models. Cross-validation results are useful for model comparison, while final performance should be assessed separately on the held-out test set.

5. HELD-OUT TEST RESULTS

Evaluation measure	Result
Baseline OLS test R^2	0.7838
Final Random Forest test R^2	0.8632
Final Random Forest test RMSE	$4,608.29

On the held-out test set of 268 observations, the final Random Forest achieved an R^2 of 0.8632 and an RMSE of $4,608.29. Compared with the baseline OLS test R^2 of 0.7838, this is an increase of 0.0794 R^2 points. The RMSE expresses the scale of prediction error in the outcome's units, with larger errors receiving greater weight.

6. DISCUSSION AND CONCLUSION

This analysis demonstrates the value of combining model diagnostics with structured model comparison. The baseline linear model provided a useful benchmark, but its residual plots suggested that a simple additive specification did not capture all patterns in the data. Adding squared and interaction features improved the cross-validated performance of the linear model, and the Random Forest achieved the strongest reported results among the models compared.

- **Diagnostics guide model development.** Residual plots and the normality test helped identify limitations in the baseline fit and motivated a more flexible specification.
- **Interactions can represent non-additive relationships.** Explicitly modeling age- and BMI-related differences by smoking status allowed the expanded linear model to represent relationships that a main-effects-only model cannot.
- **Flexible models can improve prediction.** The Random Forest produced the highest reported cross-validation and held-out test R^2 values in this comparison.

Limitations and interpretation

These findings are predictive, not causal. The analysis uses one dataset and one train/test split; performance may vary with different samples or validation schemes. Residual non-normality does not necessarily invalidate prediction, though it can affect classical inferential procedures. Further work could include repeated cross-validation, uncertainty intervals for performance metrics, calibration checks, and a clearer comparison of error measures across models. Any causal interpretation of smoking, BMI, or age would require a separate design and assumptions beyond this predictive analysis.

SUMMARY OF REPORTED RESULTS

Training sample	Test sample	Best reported model
1,070 observations	268 observations	Random Forest Regressor
CV R^2	**Test R^2**	**Test RMSE**
0.8569	0.8632	$4,608.29